









Validating Causality

50



PO

po



PO

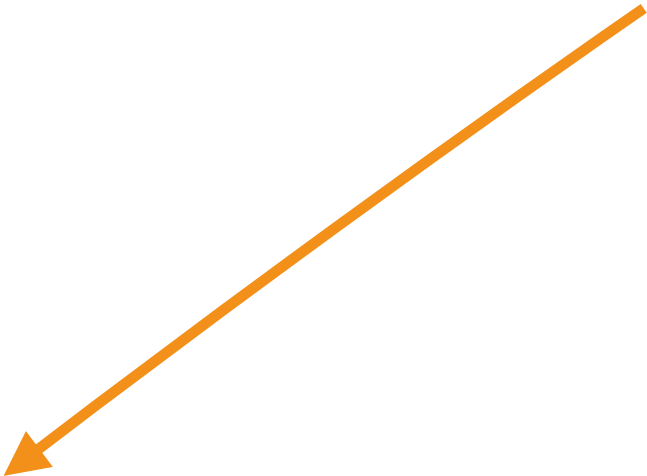
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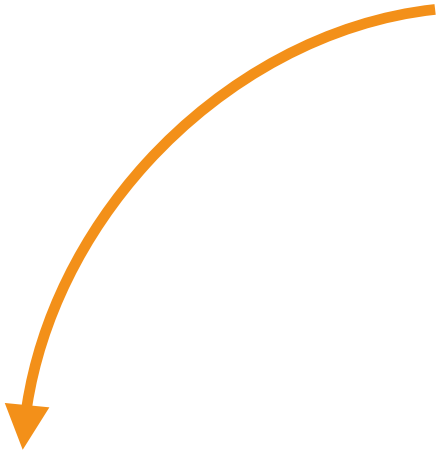


po

po



COOL ACC



collog



S



—

R  +

R



-



S



-



C



-

R  +

R.B. -



C



—

```
update(src, +100)
```

```
update(src, -100)
```

```
def update(acc: cown[Account], amount: int) {  
  when(acc) {    
    acc.balance += amount  
    val id = acc.id()  
    when(log) {  log.record(id, amount) }  
  }  
}
```

What are given the programs events

We are giving the program and you orders

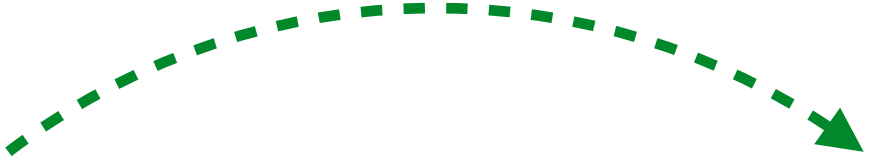
Wendyriverthunderers

We test for a cyclicity

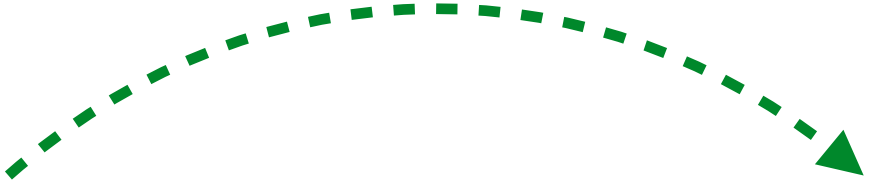




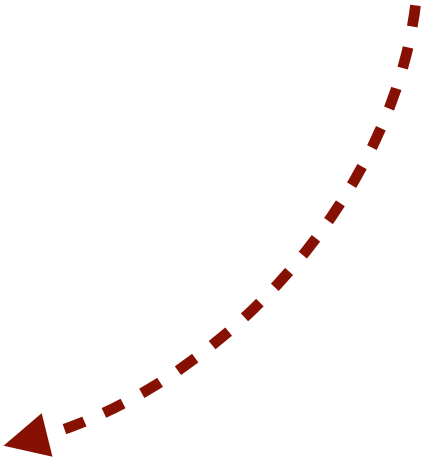












h

b



We derive the happens before orders







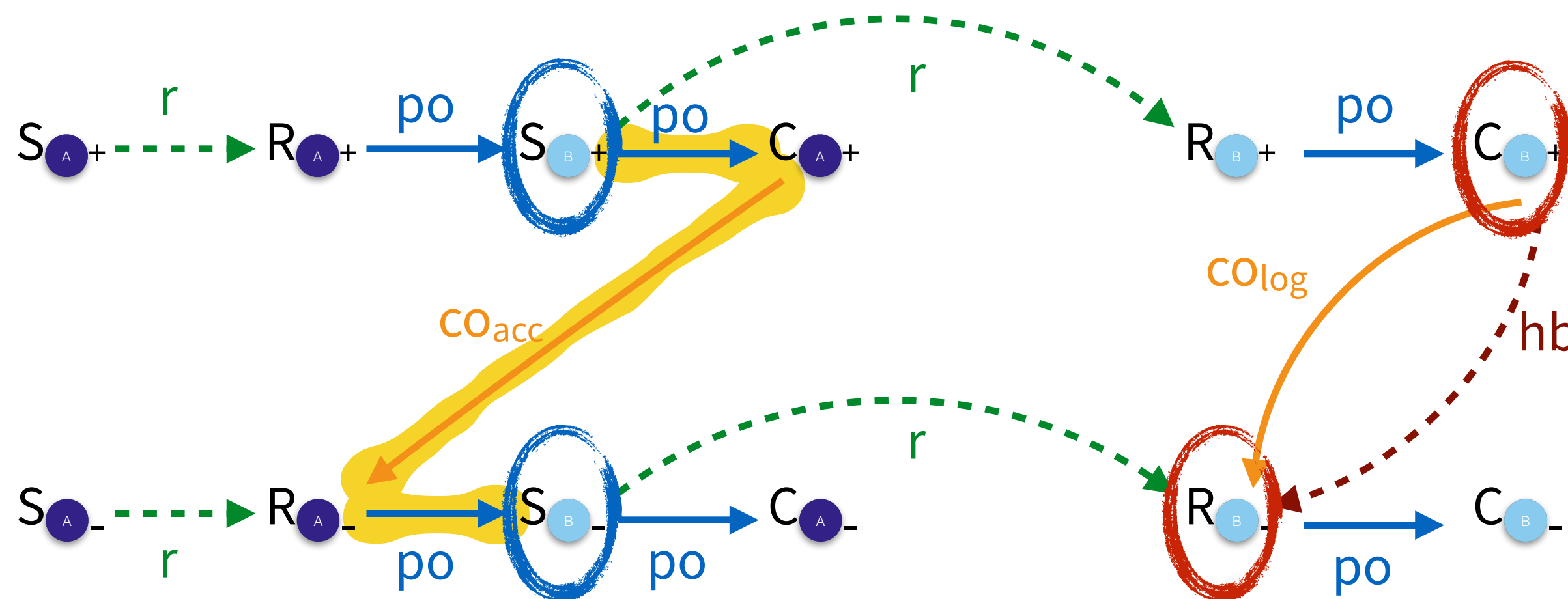


Validating Causality

```
def update(acc: cown[Account], amount: int) {  
  when(acc) { A  
    acc.balance += amount  
    val id = acc.id()  
    when(log) { B log.record(id, amount) }  
  }  
}
```

update(src, +100)

update(src, -100)



We are given the program events

We are given the program and cown orders

We derive the run orders

We derive the happens before orders

We test for acyclicity ✓

Validating Causality

```
def update(acc: cown[Account], amount: int) {  
  when(acc) { A  
    acc.balance += amount  
    val id = acc.id()  
    when(log) { B log.record(id, amount) }  
  }  
}
```

update(src, +100)

update(src, -100)

